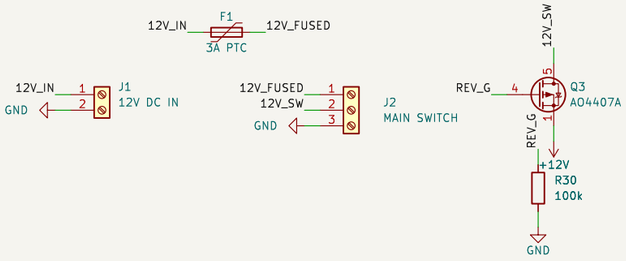
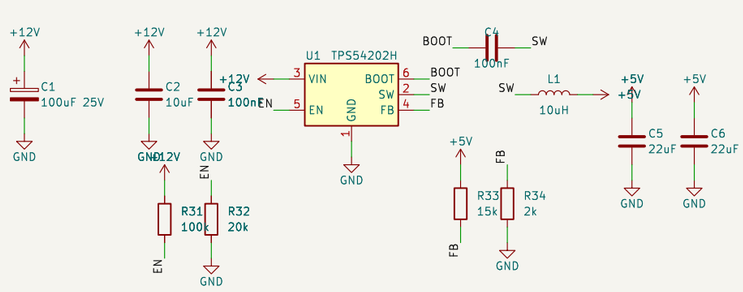
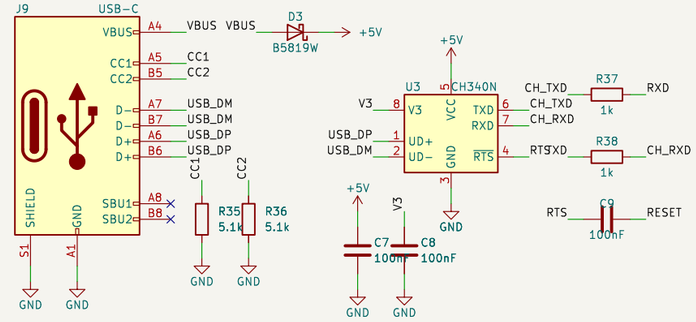
POWER IN: RESETTABLE FUSE, MAIN SWITCH, REVERSE-POLARITY MOSFET



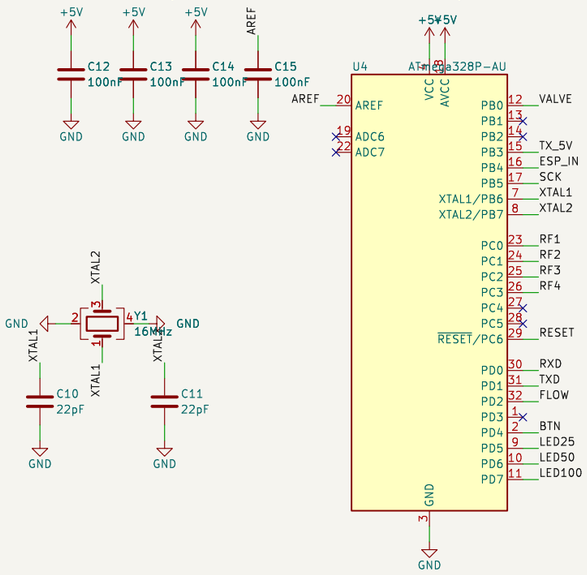
5 V BUCK REGULATOR



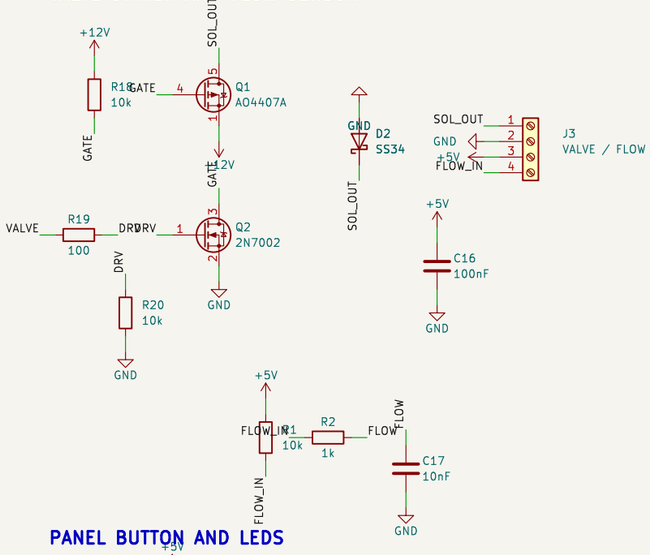
USB-C AND USB-SERIAL (CH340N)



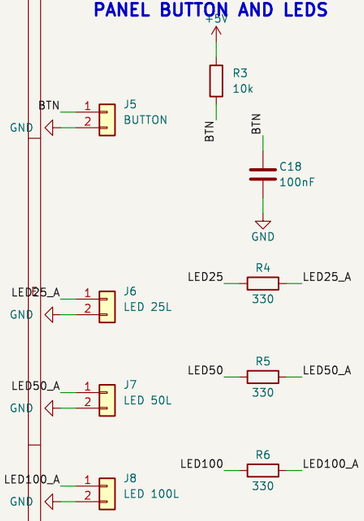
ATMEGA328P (ARDUINO UNO / NANO PINOUT)



VALVE DRIVER AND FLOW SENSOR



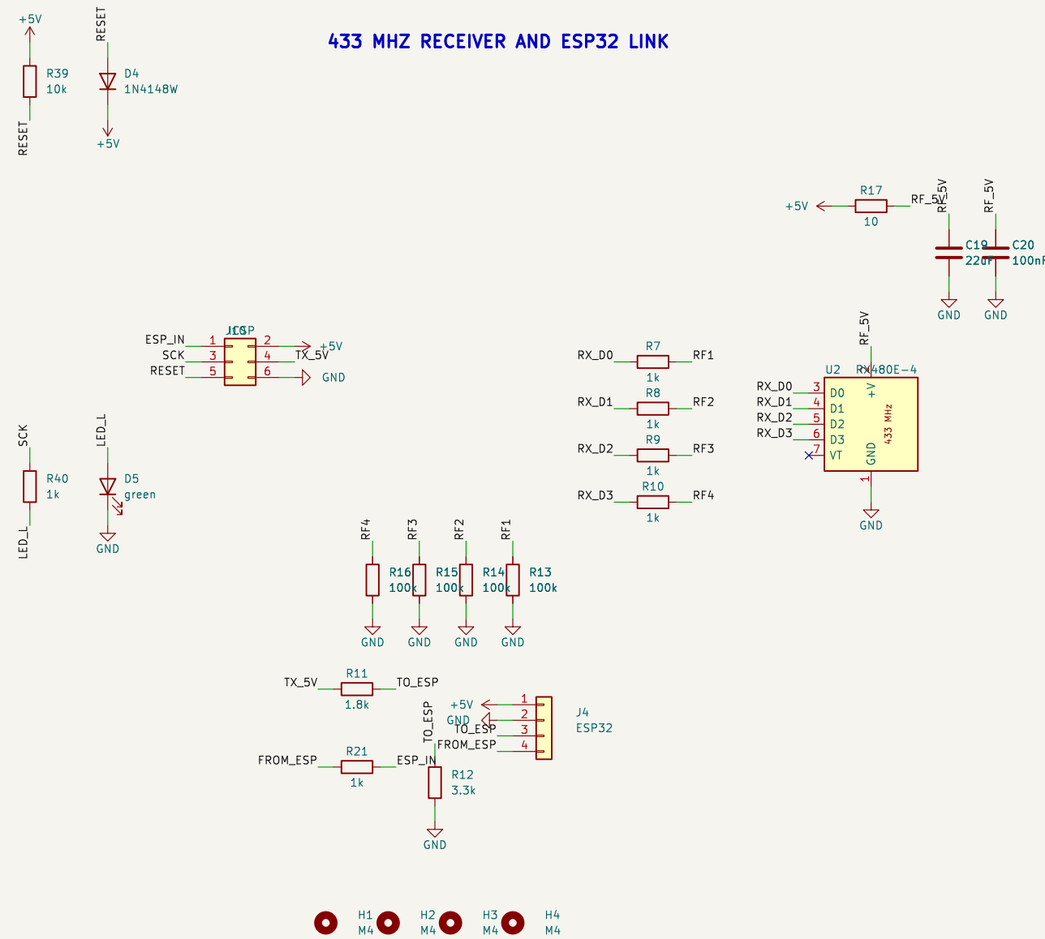
PANEL BUTTON AND LEDS



Arduino pins: D0/D1 serial (USB), D2 flow, D4 button, D5–D7 LEDs, D8 valve, D11 data to the ESP, D12 data from the ESP, D13 LED, A0–A3 remote. Same pins as the Nano boards, so WaterValveController.ino runs unchanged. Burn the bootloader once through J10 (ICSP) as "Arduino Uno", then upload over USB.

Off-board wiring: J1 12 V supply; J2 illuminated rocker switch (1 power, 2 load, 3 lamp GND); J3 valve (+ to 1, – to 2) and YF-G1 sensor (black 2, red 3, yellow 4); J5 button; J6–J8 plain panel LEDs; J4 ESP32-C3 (GPIO20 RX, GPIO21 TX).

433 MHZ RECEIVER AND ESP32 LINK



H1 M4 H2 M4 H3 M4 H4 M4

Pins: D2 flow, D4 button, D5–D7 LEDs, D8 valve, D11/D12 ESP, A0–A3 remote
Firmware: WaterValveController.ino (unchanged)
ATmega328P-AU 16 MHz, CH340N USB-C, TPS54202H 5 V buck, AO4407A valve switch
Single board, no Arduino Nano module

Sheet: /
File: WaterValveController-ATmega.kicad_sch

Title: Water Valve Controller – standalone ATmega328P

Size: A3	Date: 2026-10-12	**Rev: A1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1